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APPLICATIONS OF UAV HYPERSPECTRAL AND LIDAR DATA FOR THE GENERATION OF ULTRA-RESOLUTION IMAGERY OF MANGROVE FORESTS

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ABSTRACT

Mangrove forests have been suffering with the impact of anthropogenic activities (i.e. fishing, deforestation) in the past years. Imaging data with high spatial and spectral resolution is a tool to assist in the conservation management of these ecosystems, besides allowing the *in situ* characterization of mangrove trees. Systems onboard UAVs have been improving the applications of remote sensing for this purpose, delivering high resolution products with lower costs in comparison to airborne systems. The aim of this study is to investigate the applications of coupled UAV hyperspectral and LiDAR data for the generation of ultra-high-resolution images of mangrove forests. Here we present the pre-processing steps and preliminary results of the combined data.

Index Terms— Mangrove, hyperspectral, LiDAR

1. INTRODUCTION

Mangrove forests are complex coastal ecosystems. Apart from rich biodiversity, they also provide fundamental societal and economic services [1]. Brazil hosts the largest continuous extent of mangrove in the word (~6,800 km) [2]. Despite their importance, mangrove forests have been widely affected by anthropogenic activities, especially fishery and deforestation [3]. Monitoring mangrove forest is essential to identify the impacts of human activities and preserve these ecosystems. Remote sensing technologies have been used for the past years for this purpose [4]. Even though satellite data can be used for the estimate of mangrove extents [5], higher resolution is needed for tree-scale monitoring. Imaging systems onboard unmanned aerial vehicles (UAVs) deliver products with the ultra-high spatial and spectral resolution needed for conservation management, allowing monitoring mangrove attributes such as species biodiversity, forest health, vertical forest structure and carbon fluxes [4]. Referring to this, the aim of this study is to investigate the applications of hyperspectral and LiDAR products acquired from sensors onboard UAVs for the generation of unprecedent ultra-resolution images over mangrove forests.

2. MATERIALS & METHODS

2.1. Study site

The UAV data acquisition was carried over a mangrove forest located in Bertioga city (23°54'09.3"S, 46°12'18.4"W)). The city is located at the north coast of the São Paulo state, Brazil. The forest is composed of varied mangrove species at different stages of development, varying from seedlings to ~15 m high trees.

2.2. UAV systems

The UAV data was acquired in early August 2021 in two separate flights, one with the HySpex NEO Mjolnir VS-620 hyperspectral camera and the other with a RIGEL MiniVUX-1 UAV laser scanner. Both sensors operate in multi-copter platforms. The VS-620 consists of two pushbroom imagers (i.e. V-1240 and S-620), which combined operates with 490 bands spanning from 400 – 2500 nm range [6]. A data acquisition unit (DAU) and navigation system (Applanix APX-15 internal measurement unit - IMU) complete the system onboard the camera, allowing direct georeferencing of the imagery.

The MiniVUX-1 UAV LiDAR system operates with a 100 kHz pulse repetition rate and has a rotating mirror that provides 360-degrees line scans. The MiniVUX-1 UAV has also an integrated Applanix APX-15 IMU. Both flights were performed at 40 m above the take off point, with 2 m/s flight speed. For the hyperspectral data, the UAV survey was performed in three parallel flight lines (4500 m²), with 18 m spacing and about 25% horizontal overlap. The flight with the MiniVUX-1 UAV was also performed in three parallel flight lines, which resulted in a point cloud with a density of 700 points per square meter in average, covering an area of 3 ha (30.000 m²).

2.3. Data processing – MiniVUX-1UAV

The processing chain of the data acquired with the MiniVUX starts with the georeferencing of the UAV's GNSS/INS data at POSPac (Applanix). Here the navigation data of the UAV is optimized using a GNSS base station to assign a geographical location to the 3D coordinates of the

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point cloud generated with the LiDAR sensor. Next, the point cloud is processed with the RiPROCESS software package (RIEGL). First, the program uses the GeoSysManager database to adjust the coordinate system of the point cloud. Then, the trajectory (*i.e.* georeferenced navigation data generated at POSPac) is imported and the extraction parameters are set to generate the georeferenced point cloud. The RiPRECISION tool is used to enhance precision and increase the georeferencing accuracy of the point cloud. Once configured, the point cloud is filtered to exclude outliers, and then exported. This file in used to generate a high-resolution digital surface model (DSM) in ENVI.

2.4. Data processing – Mjolnir VS-620

The Mjolnir VS-620 generate separate images for the VNIR and SWIR flight lines, therefore the images generated for each range are processed individually and merged later. The processing workflow of the hyperspectral image (HSI) also starts with the UAV's georeferencing using the POSPac-UAV software. The HySpex NAV software (HySpex NEO) uses the data generated from POSPac-UAV to produce INS data for each frame of the flight lines. Radiance data is generated using the HySpex RAD software (HySpex NEO), in units of $W/m^2/sr/nm$. The orthorectified image is generated in the PARGE software (ReSe Applications). The program combines HySpex files (*i.e.* sensor model, boresight offsets), the outputs from HySpex NAV and the DSM (generated from the MiniVUX-1UAV point cloud), to georeference and orthorectify the at-sensor radiance data produced with the HySpex RAD. Finally, PARGE is used to merge the VNIR and SWIR flight lines into a single full range image [6]. The atmospheric correction was performed using the DROACOR (ReSe Applications), which is a software package exclusive for reflectance retrieval from UAV-based imaging data, based on the radiation parameters from the LibRadtran radiative transfer model [7].

3. RESULTS

The DSM generated from the MiniVUX1-UAV is presented in Figure 1. The model was generated with point density (points/m²) of 500 and 5 cm resolution. The HSI generated from the 2 flight lines acquired with the Mjolnir VS-620 camera is presented in Figure 2. The image has a pixel size of 5 cm.

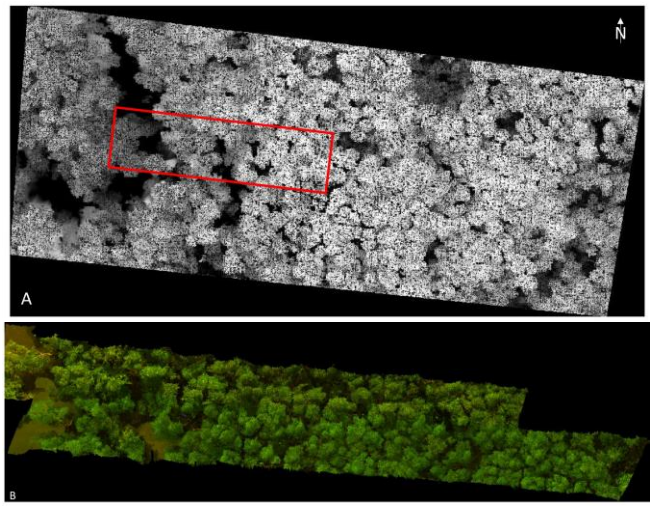

Figure 1. A. Digital surface model generated from MiniVUS1-UAV data. B. Hyperspectral image (reflectance) generated from the Mjolnir VS-620 data. The red square in the DSM delimitates the area covered by the HSI.

4. CONCLUSIONS

Here we presented the preliminary results of a study aiming for the characterization of mangrove forests with UAV based hyperspectral camera and laser scanner. The high-resolution DSM generated with the MiniVUX (LiDAR) allows the generation of a georeferenced HSI with high spectral (490 bands) and spatial (5 cm) resolution with the data acquired with the HySpex Mjolnir VS-620 camera. Further processing will be applied to these pre-processed data to conclude the processing chain and spectral characterization of mangrove forests.

5. ACKNOLEDGMENTS

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